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Start-up shear flow of a well-characterized entangled H-polymer

Frank Snijkers [a], Hyojoon Lee [b], Taihyun Chang [b], Chinmay Das [c], Dimitris Vlassopoulos [a,d,*]

[a] Institute of Electronic Structure & Laser, FORTH, Heraklion 70013, Greece
[b] Department of Chemistry and Division of Advanced Materials Science, Pohang, University of Science and Technology (POSTECH), Pohang 37673, South Korea
[c] School of Mathematics, University of Leeds, Leeds LS2 9JT, United Kingdom
[d] Department of Materials Science & Technology, University of Crete, Heraklion 70013, Greece

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ABSTRACT

We present a systematic investigation of the linear viscoelasticity and nonlinear start-up shear flow response of a well-characterized H-polystyrene, an archetype branched polymer. The Branch-on-Branch (BoB) model describes the linear response well, as well as the nonlinear transient response at lower rates, in the regime where finite extensibility effects do not seem to be important. The data reveal shear thinning similar to that of comb polymers and a broad transient overshoot of the shear stress growth coefficient, which suggests a coupled response of backbone and branches. The rich features of the data and comparison with the BoB-model should provide a useful framework for further exploring the transient molecular response of branched polymers in complex flows.

1. Introduction

Among the greatest advances in polymer science has undoubtedly been the progress in the fundamental understanding of the rheology of entangled branched polymers [1–5]. The coarse-grained mesoscopic tube model provides the basic framework for interpreting physically the viscoelastic response of these materials in the linear regime and, importantly, in the nonlinear regime, in relation to their processing performance. This progress is due to the ability to work with well-characterized polymers, typically synthesized with controlled polymerization methodologies, which allow associating a particular response to molecular features (6–14]. This knowledge allows formulation of models with specific physical mechanisms that can be further tested against commercial polymers which are polydisperse in size and molecular structure [2,15].

The key insight that led to the advances in the field of branched polymer rheology is the realization that such complex polymers, relax their stress hierarchically, once a strain is imposed, from the outer to inner regions [3,8,10,14,16,17]. The relaxed branches act as effective solvents for the inner unrelaxed parts of the molecule (i.e. for the backbone in combs or H-polymers, or inner generations in dendritically branched macromolecules). Then, the relaxation proceeds further. For example, the backbone relaxes its stress like a linear polymer in a dilated tube with increased friction coming from the branch points [14,16]. This overall hierarchical relaxation mechanism has proved to be successful for polymers of varying architectures such as asymmetric star polymers [19], Cayley-trees [20,21], combs [8–10,17,18,22], and star-combs [23].

The nonlinear response of branched polymers is complex and its understanding reasonably well developed. Typically, these polymers are known to exhibit strain hardening in extensional flows, viscosity thinning in shear flows and time-strain separability with weaker damping compared to linear polymers in step shear deformation [2,4,11–13,22,24]. These features can even be modeled nearly quantitatively with the help of models of different level of coarse graining and sophistication [1–4,25–30]. This level of understanding has been achieved with the help of model (well-characterized) polymers, obtained by combining state-of-the art synthesis and advanced characterization [6,7]. Using such materials it has been possible to decode their complex response in different flow conditions (steady, transient, oscillatory, shear, extension) and explore its molecular origin [1–15]. There are however only a few studies of nonlinear response with well-characterized branched polymers like combs. For example, uniaxial extensional rheology of comb polymers has revealed the importance of dilution due to branch relaxation and withdrawal [13,25], which has led to improvement of mesoscopic models [29,30]. The shear measurements are particularly difficult because rotational rheometry is prone to artifacts. With the help of recent developments, such as the cone-partitioned plate (CPP) tool, the edge fracture instability is mitigated and reliable measurements can be performed [31–36]. A systematic study of comb

* Corresponding author.
E-mail address: dvlasso@iesl.forth.gr (D. Vlassopoulos).

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polymers has revealed their weaker shear thinning compared to linear entangled polymers and the unusual transient response, manifested as two local overshoots in the shear stress growth coefficient [11,12]. They were linked to the response of the branches and the dilute backbone once the branches were withdrawn into the backbone's tube. This latter point, the fundamental understanding of the interplay of branches and backbone during shearing requires further investigations with other well-defined branched macromolecules. To this end, it is important to consider the relatively simple H-polymer architecture, which is a simplified version of comb, with only two branches grafted at the two ends of a backbone, whoseo linear rheology and structure are established [14]. It is especially interesting to consider the situation of a H-polymer whose branches and backbone have nearly the same molar mass (one of the situations where there is a marked departure from star polymers). Focusing on transient response is important both for polymer physics (as it provides insights into the evolution of the material's response to steady state via different molecular mechanisms such as orientation and stretching) [1,2,11,12] and technological applications [2,4,15,28]. The latter relate mainly to processing because knowledge of the molecular mechanisms involved in the flow of branched polymers and associated with branches or backbone can help in optimizing processing conditions, whereas in addition, the time dependence of different material functions is important to monitor and/or improve material performance and stability [4,15].

Motivated from the above challenges and the need to provide additional systematic experimental evidence of the response of such model materials, in this work we embarked on an investigation of a well-characterized H-polymer with branches and backbone having nearly the same molar mass. In such a case, the dilution effect of the branches is expected to be significant. We present below the characterization (which confirms the structural purity of our material) and the shear rheology (linear and nonlinear), along with analysis, and compare with the results from comb polymers.

2. Materials and methods

2.1. Molecular characteristics of the H-polymer

A general schematic of an H-polymer is shown in Fig. 1. Its structure is in principle very well defined, as the number of branches [4] and their exact position (2 on each end) are exactly known. The H-polymer under consideration here is a polystyrene sample synthesized using high-vacuum anionic polymerization and characterized by J. Roovers in the 80 s [37,38].

Table 1 lists the molecular characteristics of the H-polymer named H3A1A [37,38]. The molar mass between entanglements M_e (in kg/mol) in Table 1 is that used typically in the literature [39] and consistent with the experimental data, since [4] $M_e = \frac{\rho R(T+273.2)}{G_N^0}$ with G_N^0 the plateau modulus, ρ the mass density of the polymer (in kg/m^3; at 169.5 °C, ρ = 980 kg/m^3) [40], R the universal gas constant, and T the temperature (in °C). Table 1 further displays the weight-averaged molar mass of the total H-polymer M_{TOT}, the backbone M_{BB} and a branch M_{Br} (in kg/mol). A ratio of M_{Br}/M_{BB} equal to one was targeted in [37] for the synthesized series of H-polymers. Note that the simple summation $M_{TOT} = 4.M_{Br} + M_{BB}$ is not an identity because the addition of reactants to obtain the

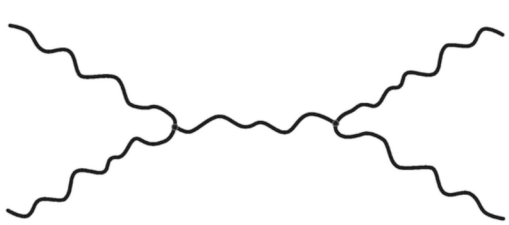

Fig. 1. Schematic of an H-polymer.

Table 1
Molecular characteristics. See text for definitions of the quantities. Molar masses are in kg/mol.

SAMPLE	M_e	M_{TOT}	M_{BB}	Φ_{Br}	$Z_{BB}^{DIL.}$	M_{Br}	Z_{Br}
H3A1A	17	674	123	0.78	5.6	132	7.7

final H-polymer from the linear backbone and branches can lead to some side-reactions, and due to experimental errors in the molar mass determination [37]. Finally, Table 1 lists the number of entanglements of a branch $Z_{Br} = M_{Br}/M_e$, the volume fraction of the branches $\Phi_{Br} = 4.M_{Br}/M_{TOT}$ in the total H-polymer and the number of entanglements of the dynamically diluted backbone (i.e. treating the branches as solvent for the backbone and assuming a dynamic dilution exponent α equal to 1) defined as $Z_{BB}^{DIL.} = \phi_{BB}^{\alpha}.Z_{BB}$.

2.2. Shear rheological measurements with a cone-partitioned plate (CPP)

We used a strain-controlled ARES rheometer (TA Instruments) equipped with a force rebalance transducer (2KFRTN1) and air/nitrogen convection oven with a temperature control of ± 0.1 °C (a controlled environment of nitrogen was employed to reduce the risk of degradation). The study of nonlinear flow properties of polymer melts on a rotational rheometer is complicated by the occurrence of flow instabilities and other artifacts such as edge fracture. Following the pioneering work of Meissner and co-workers [31] and the subsequent extensive investigations by Schweizer and co-workers [32–34], we constructed a CPP geometry with temperature control for the strain-controlled ARES rheometer [35,36]. It allows performing nonlinear flow measurements unaffected by edge fracture up to higher strains and shear rates as compared to a regular cone and plate geometry (eventually, also for the CPP set-up, edge fracture takes over, but the effects are delayed). The technical realization and performance of our simple CPP on linear monodisperse entangled polymers (PI and PS) was investigated in Ref. [35] and here, the same CPP set-up is used to investigate the H-polymer.

2.3. Temperature gradient interaction chromatography (TGIC)

TGIC analysis was performed in order to check the structural purity of the sample [41,42]. The TGIC chromatrogram of H3A1A (Fig. 2) shows a well-discerned peak and confirms the good quality of the

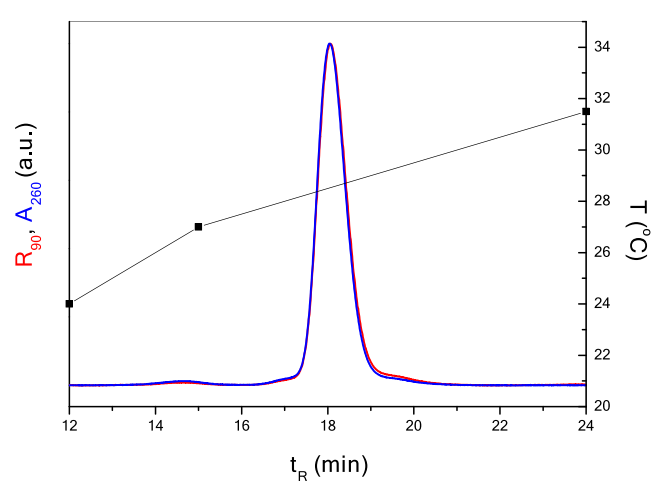

Fig. 2. Chromatogram from TGIC analysis of H3A1A. The signals of UV adsorption (blue) and light scattering detector (red) are shown for a given temperature gradient (data) over the elusion time. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

polymer in terms of dispersity. Absence of any additional peaks reveals lack of any degradation or crosslinking since synthesis. The well-matched signals of the elusion peak from light scattering and UV adsorption confirm the very good structural purity of the sample. Additional evidence (in particular concerning the absence of higher molar mass species) comes from the excellent comparison of the linear viscoelastic spectra (old and new data), and in particular the sensitive loss factor tan(δ) (see Fig. 3 below).

3. Experimental results

3.1. Linear viscoelasticity

The linear rheological data of H3A1A is shown in Fig. 3 as master curves for the dynamic storage G' and loss G'' moduli and their ratio tan (δ) = G′/G′ (known as loss factor) as a function of angular frequency ω at a reference temperature T_{ref} of 169.5 °C, as taken from [38]. Note that the relaxation time of the branches is relatively close to the terminal relaxation time and that, after the relaxation of the branches, the typical secondary plateau for the backbone is very narrow (nearly absent). This is not unlike the situation for the polystyrene combs with the longer branches in Refs. [8,42], especially for the combs denoted c642 and c652.

Table 2 lists an overview of the dynamic properties of the H-polymer as obtained from the linear data in Fig. 3. The value for the plateau modulus G_N^0 is taken as value of the storage modulus G' at the minimum of the phase angle tan(δ) = G′/G'. The zero-shear viscosity $\eta_0 = \lim_{\omega\to 0}\frac{G'}{\omega}$ and the zero-shear recoverable compliance $J_S^0 = \frac{1}{\eta_0^2}\lim_{\omega\to 0}\frac{G'}{\omega^2}$ are calculated from the linear oscillatory data. Their values and estimated errors are obtained as described in Ref. [12]. The longest relaxation time τ_0 listed in Table 2 was calculated as $\tau_0 = J_S^0.\eta_0$. It is used further on to define a characteristic Weissenberg number $Wi_0 = \tau_0.\dot{\gamma}$, with $\dot{\gamma}$ the shear rate.

The Branch-on-Branch model (version 2.6) [1,2] was used to attempt to model the linear oscillatory data and the (Fig. 3) using the following input: a molar mass between entanglements M_e of 16.6 kg/mol, a density ρ of 1000 kg/m^3, a reference temperature T of 170 °C and an entanglement relaxation time τ_e of 0.001 s. The model parameters, the dilution exponent α, and, the branchpoint hopping parameter p^2 were further selected to be 1 and 1/40, respectively, following Ref. [1]. All other choices for the relaxation were set to their default values. For the polymer structures, the molar masses for the branches and backbone

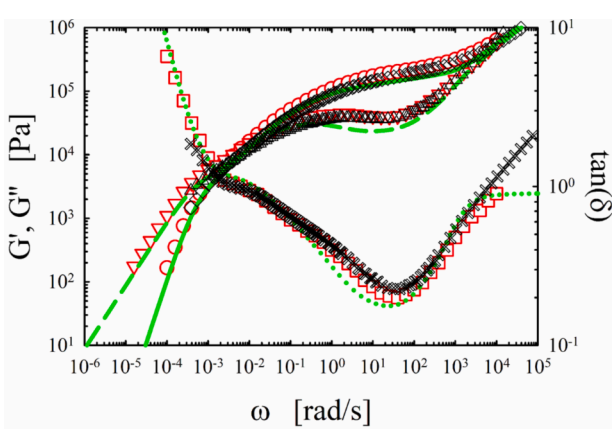

Fig. 3. Master curves for the dynamic moduli G', G″, and tan(δ) as function of angular frequency ω at a reference temperature of 169.5 °C for the H3A1A. The old data from Ref. (38) (red symbols: G' - circles, G″ - downward triangles, and tan(δ) - squares) is compared to newly collected data (42) (black symbols: G' – diamonds, G″ - upward triangles, and tan(δ) – crosses) and display excellent agreement. Predictions of the BoB-model are also shown (green lines: G' - full line, G″ - dashed line, and tan(δ) - dotted line).

Table 2
Viscoelastic properties of H3A1A (see text for definitions).

SAMPLE	$T_{ref.}$ (°C)	$G_N^0/10^5$ (Pa)	$\eta_0/10^5$ (Pa.s)	$J_S^0/10^{-5}$ (Pa^{-1})	τ_0 (s)
H3A1A	169.5	2.0	113 ± 3	14 ± 4	1600 ± 400

from Table 1 were used assuming a lognormal molar mass distribution with a polydispersity value of 1.05 for both branches and backbone. Whereas the polydispersity of the final fractionated product (as well as the branch and backbone) is not reported in the synthesis paper, it is emphasized that it is very narrow [17], as expected for high vacuum anionic synthesis. Based on this and the chromatogram of Fig. 2, the value of 1.05 used in the model is very reasonable. Comparison of the model and the experimental data is shown in Fig. 3, where the predictions are shown in green. The comparison is excellent, except for the small discrepancy for the plateau modulus, which is not uncommon for the BoB-model [1,44]. The plateau modulus, calculated as $G_N^0 = \frac{4}{5}\frac{\rho R(T+273)}{M_e}$, is 0.18 MPa. One possible way to match the experimental data better will be to relax the relationship between G_N^0 and M_e, by choosing a different prefactor compared to the rubber theory prediction 4/5. It should also be noted that the value of M_e (slightly different in the model compared to Table 1) dictates the width of the relaxation and to some extent depends on the model to describe the relaxation. As can be seen in the tan(δ) plot of Fig. 3, the chosen value captures both the relaxation of the branches and the final reptation of the backbone.

3.2. Nonlinear start-up shear flow

The shear stress growth coefficient (transient start-up shear viscosity) η^+ is shown as function of time for different shear rates in Fig. 4 at a reference temperature of 169.5 °C. The thick dark grey line is the linear viscoelastic (LVE) limiting line calculated from the linear data of Fig. 3 (Cox-Merz and Gleissle rules). Start-up shear experiments were performed on two separate sample loadings, one at 169.5 °C (light grey and black lines) and another at 200 °C (thin dark grey lines). The data from the experiments at 200 °C are shifted to 169.5 °C using the time–temperature shift factors with the horizontal shift factors following WLF and the vertical shift factors following the density-temperature scaling. After the standard TTS shifting, the data obtained at 200 °C was still clearly too high compared to both the LVE-line and the nonlinear data obtained at 169.5 °C. We can attribute this discrepancy to different reasons, the most obvious being errors in the shift factors and overloading effects in the CPP, which are not always negligible. The data from 200 °C was hence shifted downward with an artificial factor of 1.10. The rates quoted in the legend are TTS shifted rates in s^{-1}, at 169.5 °C. In Fig 4a the experimental nonlinear start-up shear data are presented, and the evolution of overshoot and undershoot in the shear stress growth coefficient (also called transient start-up viscosity) as the shear rate increases is clearly observed. Selected data (black lines) and predictions of the BoB-model (grey lines) are compared in in Fig. 4b. Shear rates are mentioned next to the lines in s^{-1}, the thick lines are the LVE lines for both experimental data (black) and predictions (grey). In depth comparison between model and data will be analysed and discussed further below, but one can observe immediately that the predictions for the locations (time or strain) for the maxima are rather comparable to the experimental data, while on the other hand, the predictions do have a much steeper/pronounced overshoot with an overall stronger thinning behavior for the steady state viscosities. We estimate the Rouse time of the H polymer as ~ 0.9 s from $\tau_R = Z_{Tot}Z_{Span}\tau_e$ [13] with $Z_{Span} \equiv Z_{BB} + 2Z_R$ being the span backbone molar mass in units of M_e. Finite extensibility effects, absent in the pompom model implemented in the BoB-model, can have significant effects on the nonlinear response at rates exceeding $1/\tau_R$. The predictions show that at rates exceeding 1 s^{-1}, the model underpredicts the stress maximum and overpredicts shear

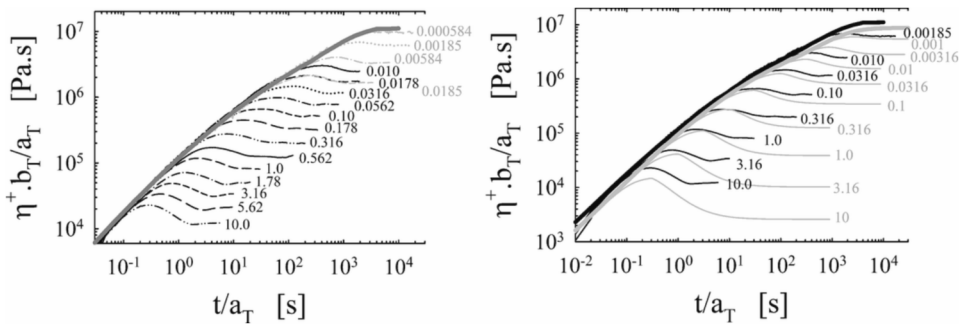

Fig. 4. (a) Transient start-up shear viscosity η^+ as function of time for different shear rates at a reference temperature of 169.5 °C. Shifted shear rates at 169.5 °C are shown next to the lines. The black lines represent data directly collected at 169.5 °C, the light grey lines represent data collected at 200 °C and shifted to 169.5 °C. (b) Selected experimental nonlinear start-up shear viscosity as in Fig. 4 (black lines) and the predictions of the BoB-model (grey lines). Shear rates are mentioned next to the lines in s^{-1}, the thick lines are the LVE lines for both experimental data (black) and predictions (grey).

thinning.

4. Further analysis and discussion

Fig. 5 depicts further analysis of the nonlinear start-up shear data of the H3A1A and comparisons with the BoB-model predictions and the literature data from Refs. [11] and [35]. The scaled steady-state viscosity η_{STEADY}/η_0 (Fig. 5a and b), the height η_{MAX}/η_{STEADY} (Fig. 5c and d) and the position γ_{MAX} (Fig. 5e and f) of the overshoot are shown as function of the characteristic Weissenberg numbers (based on τ_0). In Fig. 5a, c, and e a comparison of the data for the H-polymer is made with the data for a linear, monodisperse PS 182 kg/mol [35] (blue diamonds) and the PS c642 comb [11,12] (red triangles). Further, in Fig. 5a the different thinning regimes with their respective slopes are shown. At first sight, a thinning slope of about −0.9 reflects the shear thinning of entangled networks, while a slope of about −0.5 is associated with the thinning of short unentangled chains [4,5]. The investigated sample has branched segments (Table 1) which seem to be responsible for the slope at high rates. On the other hand, at lower rates, keeping analogy with the linear response (Fig. 3) the backbone has been diluted and nearly all entanglements are lost. The same behavior has been observed for comb polymers with similar molecular characteristics and dilution effects [12]. Fig. 5b, d, and f show a comparison of the data for the H3A1A H-polymer (black circles) with the predictions from the BoB-model (green squares). In Fig. 5a and b, the symbols represent the data obtained from the nonlinear step-rate measurements/predictions, the lines are obtained from the linear oscillatory data by dividing the complex viscosity η^* by η_0 and plotting this quantity as a function of the respective Deborah number $De_0 = \omega_0\tau_0$, following the Cox-Merz rule [43]. Interestingly, the empirical Cox-Merz rule seems to be followed rather well in all cases as in Ref. [44]. An interesting trend can be observed in Fig. 5c for the position of the strain where the maximum occurs. At $Wi_0 < 1$, there is no overshoot (linear regime). At higher rates, $1 < Wi_0 < \sim 100$, orientation and then stretch of the backbone seems to occur while at even higher rates, $\sim 100 < Wi_0 < 10000$, the position of the overshoot decreases again and seems to be taken over by the orientation of the branches (peak position is at a strain of ~ 2). At the highest rates, $Wi_0 > 10000$, the position of the maximum increases again indicating possible stretch of the branches. This picture seems to suggest that, although there is no second overshoot apparent in the start-up data as for the PS c642 comb [6], there are in fact distinctly different regions in the data.

As there does not appear to be a clear secondary overshoot in the data, perhaps due to the small number of segments in the polymer, while the plot of the strain at maximum viscosity (Fig. 5c) does suggest separate phenomena, a further analysis is made of the breadth (width) of the overshoot, i.e. if there are two not well-separated peaks, the breadth of the overshoot should presumably be larger than for other, e.g. linear, polymers. Fig. 6a shows the transient start-up shear viscosity scaled with peak-viscosity η^+/η_{MAX} as function of time divided by time where the maximum occurs t/t_{MAX} for different shear rates for the H3A1A at a temperature of 169.5 °C. Plotted in this way, a very clear assessment of small differences in the shape of the overshoot can be made. Upon following the trend from low to high rates, it becomes evident that the shape of the overshoot changes significantly and more importantly, along the expected lines, e.g. the overshoot at the rate of 0.1 s^{-1} does appear to display a slight overshoot right before the dominant overshoot. The strain of this event is around 1.8. This could be interpreted as a sign of branch orientation, however there is no further evidence to support this conjecture, hence we leave it as a speculative suggestion. At higher rates the peak becomes much narrower and more symmetric. Fig. 6b shows a comparison of the data for the H3A1A with comb PS c622 from Ref. [12]. The two polymers are compared at two near identical characteristic Weissenberg numbers. The PS c622 comb has relatively short branches (with molar mass below M_e), and hence its behavior is closer to that of a linear polymer. A similar observation can be made upon comparing the response of the H-polymer with the predictions from the BoB-model (green lines in Fig. 6b). The non-symmetric shape of the transient curve (which is typical for branched polymers [11,12] and the key for broadening) precludes analysis by means of fitting with a parabola. Instead, a phenomenological way to quantify the broadening is to calculate the full width at half maximum (FWHM), i.e., take the height (distance from steady state to maximum of the transient signal) and then calculate the normalized width. The latter is the extent of the curve, i.e., the difference of time at the right end (decrease of transient start-up viscosity) to the time at the left end (increase), divided by the time at maximum. This is depicted in Fig. 7 which suggests that, within the accuracy of the performed analysis, there is substantial broadening (the average value of FWHM is about 2) and, importantly, it is the same independently of Weissenberg number and of molecular structure (H-polymer experiment and modeling, comb polymer experiment), suggesting that whereas the branch points response (withdrawal in flow) is at its origin, their effect may not be cumulative (H- polymer vs comb). This however would require further analysis with more branched structures before drawing definite conclusions.

5. Conclusions

We have presented a systematic investigation of the linear and nonlinear shear rheology of a well-characterized moderately entangled H-polystyrene having branches and backbone of nearly the same molar mass. The BoB model is found to well describe the linear viscoelasticity as well as the nonlinear start-up shear response, with overshoot and steady state, at lower shear rates. It starts deviating at rates where, apparently, finite extensibility effects are important and should be considered in future improvements of the model. Therefore, further improvement of the model would be particularly useful, and in this

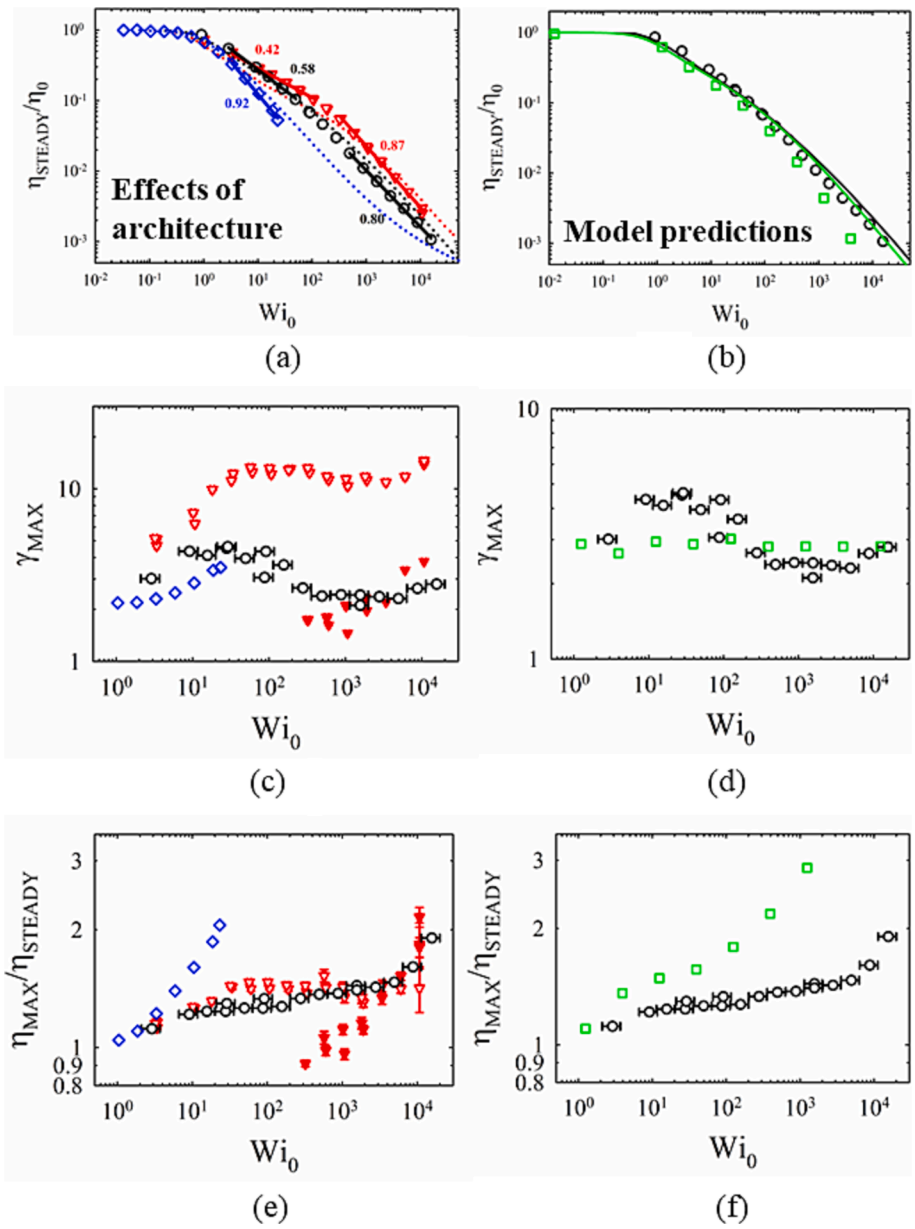

Fig. 5. In (a) and (b) the steady viscosity scaled with zero-shear viscosity η_{STEADY}/η_0, in (c) and (d) the maximum peak-viscosity scaled with steady viscosity η_{MAX}/η_{STEADY}, and in (e) and (f) the strain where the maximum occurs γ_{MAX} are shown as function of Weissenberg number Wi_0, respectively. In the left-hand panel, the data for the H3A1A H-polymer (black circles), a linear, monodisperse PS 182 k (19) (blue diamonds) and the PS c642 comb (6) (red triangles) are compared. In the right-hand panel, the data for the H3A1A H-polymer (black circles) is compared with the predictions of the BoB-model (green squares). In (a) and (b), the symbols are for the nonlinear data, and the lines for the linear data following the Cox-Merz rule (blue, red, and black dotted lines in (a) are for the linear PS 182 k, the PS c642 comb, and the H3A1A, and the full black and green lines in (b) are for the H3A1A and the BoB-model predictions). Regimes with different slopes are indicated in (a). In (c) and (e) both filled and open red triangles are shown due to the appearance of two overshoots for the PS c642 comb polymer.

context the analysis of hierarchical relaxation of H polymers with molecular dynamics simulations can provide further insights [45]. The H-polymers exhibit shear thinning viscosity similar to the comb polymers. The nonlinear data suggest a separate backbone and branch orientation or stretch, consistent with earlier findings with comb polymers, however these two mechanisms appear to be coupled. Indeed, analysis of the transient signal of the shear stress growth coefficient indicates that at a certain Weissenberg number the peak is much broader than those of the respective comb polymer with short branches but there is no clear secondary peak in the H-polymer.

H-polymers are the archetype branched polymeric structure [14], hence we believe that this systematic investigation provides a useful dataset for further analysis and essing existing models and using them to advance our understacomparisons in the direction of finnding of more complex flows of branched polymers.

CRediT authorship contribution statement

Frank Snijkers: Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Data curation. **Hyojoon Lee:** Methodology, Data curation. **Taihyun Chang:** Methodology, Formal analysis. **Chinmay Das:** Methodology, Formal analysis. **Dimitris Vlassopoulos:** Conceptualization, Writing, Methodology.

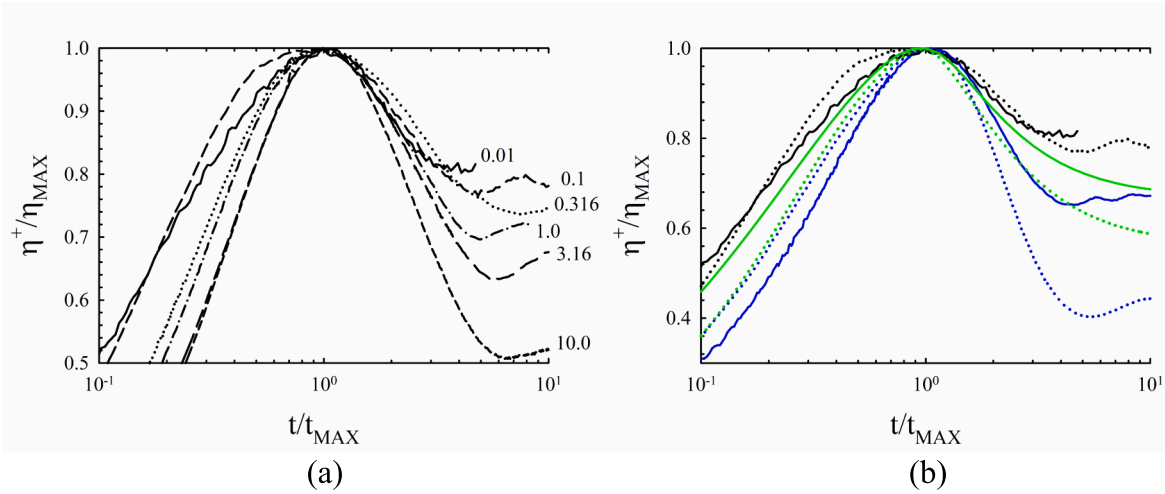

Fig. 6. Transient star-tup shear viscosity scaled with peak-viscosity η^{+}/η_{MAX} as a function of time divided by time where the maximum occurs t/t_{MAX} at a temperature of 169.5 °C. In (a) selected data is shown for the H3A1A polymer with shear rates noted next to the lines. In (b) the H3A1A polymer data (full black line $Wi_0 = 14.3$ and dotted black line $Wi_0 = 143$) is compared with data for the PS comb c622 from Ref. (12) (full blue line $Wi_0 = 15.7$ and dotted blue line $Wi_0 = 143$) and the predictions from the BoB model (full green line $Wi_0 = 12.5$ and dotted green line $Wi_0 = 125$).

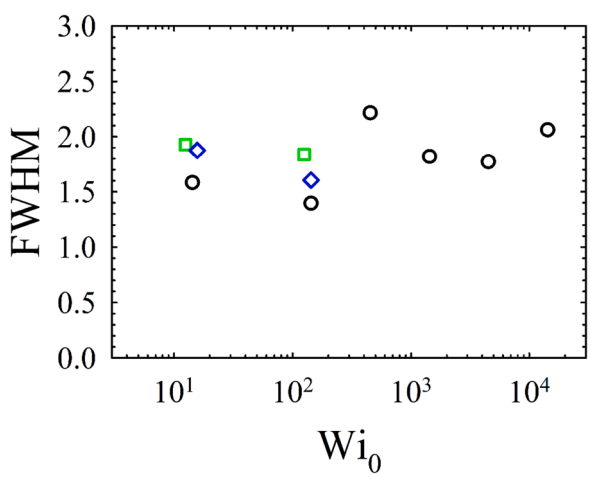

Fig. 7. Broadening analysis of the transient start-up shear viscosity signal, representing the full width at half maximum (FWHM) as a function of the Weissenberg number Wi_0. Data are shown for the H3A1A H-polymer (black circles), the PS comb c622 from Ref. (12) is compared with data for the PS comb c622 from Ref. (12) (blue diamonds) and the predictions from the BoB-model (green squares).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

It is an honor to contribute to the special issue honoring the 80th birthday of Nikos Hadjichristidis, a longtime friend and collaborator. His impact to polymer science is longlasting, and we wish him many more years of creativity and scientific leadership. We are grateful to J. Roovers, another polymer science legend, for generously providing the polystyrene H-polymer used in this work. Support for this research was received by the European Commission, FP7, ITN DYNACOP (grant 214627).

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